Crash causation with an active-inference response process: method and results

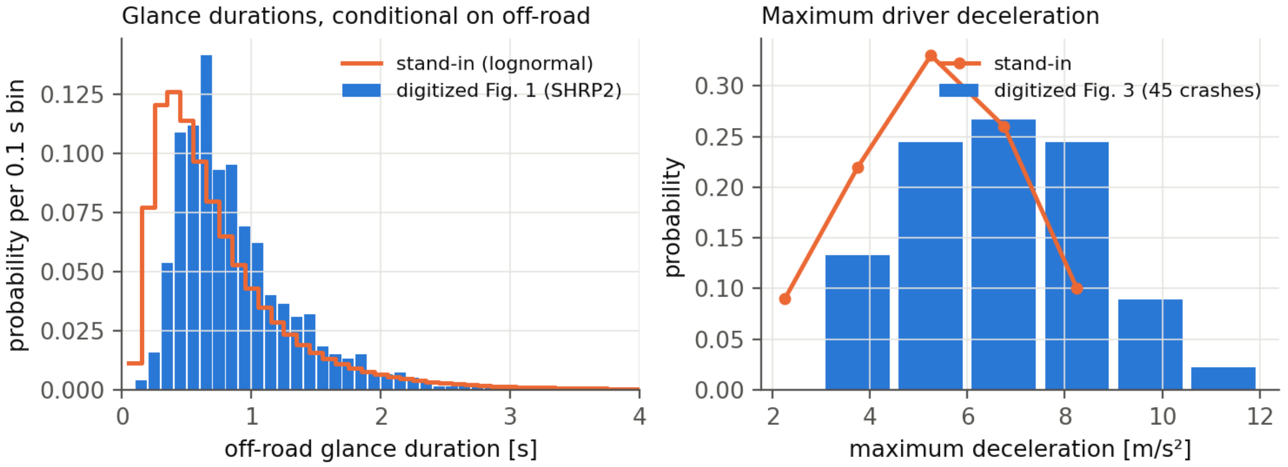
Results document, 2026-08-24 (revised the same day after the counterfactual and glance-placement decisions; the first version's numbers are superseded but reproducible from the tagged outputs). Prepared by Claude at the request of Jonas Bärgman. Markdown is the source; Word and PDF are generated from it. The design, the reading notes on the source papers, and the protocol live in `docs/crash_causation_plan.md`. Provenance tags: [B24] Bärgman, Svärd, Lundell & Hartelius (2024), [W25a] Wu et al. (2025, scenario generation), [W26] Wu et al. (2026, practical validation), [Code] the Schumann et al. released code, [OSF] their deposit, [Repo] this repository, [Opinion] a judgment.

1 The method in one page

The question: can the active-inference driver model serve as the *response process* inside a crash-causation model — and what does it add or cost relative to the fixed-delay response of the crash-causation model (CBM) of [B24]?

- **Seeds.** 100 rear-end pre-crash scenarios sampled weight-proportionally (stratified by initial speed) from the 5 000 synthetic QUADRIS scenarios of [W25a]; every seed carries its real-world weight ω.
- **The counterfactual.** Before the modeled response, the follower runs the seed's original speed profile **with its braking removed from the lead's braking onset onward** (acceleration clamped at ≥ 0). This keeps the accelerating creeping/queue seeds intact while excluding the generator-follower's own evasive action, which the plain "original" profile embeds in 14 of 100 seeds. Clamping from t = 0 instead changes headline results by only a few percent (section 4.4), so the rule choice is not consequential.
- **Causation components**, each switchable: off-road glances, too-close following (inherent in the seed), a maximum-deceleration cap, a no-response mixture (10% of crashes [B24]), and — added at the study's request — the **abnormal-acceleration follower** of [W25a]: 9.2% of crashes, the follower ignoring the lead at a constant +1.8 m/s^2 (their fitted mean; their onset-time distribution is unpublished, so the acceleration is applied from the lead's braking onset).
- **Input distributions** digitized from [B24]'s published figures (`replication/causation/digitize_b24.py`): the SHRP2 off-road glance distribution (vector-exact; two independent axis calibrations agree within 0.9%; on-road point mass 0.80) and the 45-crash maximum-deceleration histogram (integer counts summing to exactly 45).
- **Glance placement.** Primary: the **renewal process** — on/off glance sequences over the whole scenario, off-road durations from the digitized distribution, on-road dwells exponential with the mean set by the 80% on-road share, 50 Monte Carlo draws per seed (flagged for a later increase). The [B24] anchored-overshot construction is kept as the CBM-native comparison case; section 4.5 quantifies what the process placement buys.
- **Two response processes** behind one interface: the CBM's (respond 0.5 s after eyes return; ramp to the drawn deceleration) and the tier-1 active-inference surrogate (the model's preference function on the seed kinematics, fed through its evidence accumulator, drift rate calibrated on the authors' 896 deposited trials [Repo]).
- **Conditions.** A: attentive active inference (benchmark). B: active inference + all components. C: CBM response + all components (the control). D: active inference + glances only.
- **Comparison** against the full 5 000-scenario reference with the [W26] equivalence framework (θ/Θ, ROPEs 0.10/0.05, bootstrap HDIs), crashes weighted by Eq. 10, plus the exposure-weight variant (plan §6b). Metrics: P_inj, **relative speed at impact** (the assumption-free severity primitive; the deposit's delta-v was derived from it under equal masses), t_nr, and the follower's minimum acceleration.

2 Input distributions



The digitized glance distribution is close to the earlier lognormal stand-in (mode ~0.2 s later); the deceleration distribution is not — the real one has 24% of its mass at 7.5 m/s^2 and harder, where the stand-in had almost none. All results below use the digitized distributions.

3 Crash generation

With the no-brake counterfactual and process glances (the primary configuration; anchored placement for condition A, which has no glances):

condition	response	components	seeds crashing (any draw)	weighted crash prob
A	active inference	decel. cap	24/100	0.009
B	active inference	glances, decel. cap, no-response	99/100	0.023
C	CBM (control)	glances, decel. cap, no-response	56/100	0.005
D	active inference	glances, decel. cap	99/100	0.023

1. **An attentive active-inference driver avoids 76 of 100 QUADRIS crash seeds** across the full deceleration sweep; the 24 that crash in some bin are the kinematically hard core.
2. **The response process drives a factor ~4–5 in crash probability** (B 0.023 versus C 0.005 with identical components), from timing: the active-inference accumulator's attentive onsets sit at a median 1.25 s after the τ^{-1} = 0.2 s^{-1} anchor (IQR 0.50–1.75 s) against the CBM's fixed 0.50 s (section 5).
3. The crash rate was initially sensitive to the accumulator's starting level (zero start 0.023 versus 0.004 with a "stationary" half-threshold start), but the tier-2 arbiter run has **resolved the convention in favor of the zero start** (section 5): the closed-loop model's own onsets match the zero-start surrogate to a median 0.42 s across 24 seeds and reject the stationary start on 20 of 24. The 0.023 figure stands; the stationary variant remains in the outputs as a tested-and-rejected sensitivity.

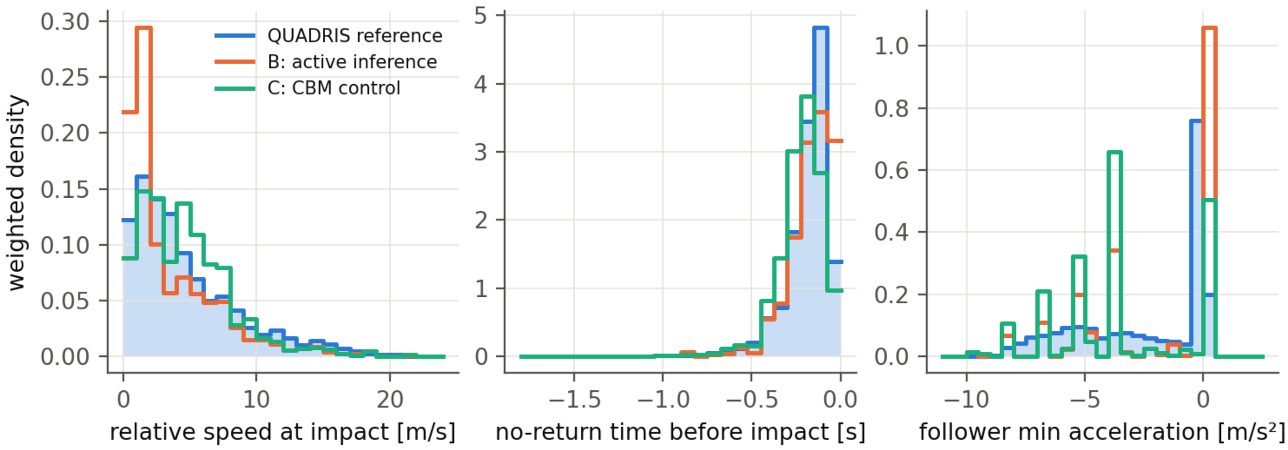
4 Comparison with the original data

4.1 Summary statistics

Weighted medians (weighted mean for P_inj); reference weighted by ω, conditions by the Eq. 10 crash weights:

	rel. speed at impact [m/s]	mean P_inj	t_nr [s]	follower a_min [m/s^2]
QUADRIS reference (n = 5 000)	3.54	0.0062	−0.15	−1.03
B: active inference, process glances	1.88	0.0048	−0.15	0.00
C: CBM, process glances	4.25	0.0055	−0.20	−3.75
B + abnormal acceleration	2.61	**0.0063**	−0.15	0.00
C + abnormal acceleration	4.41	0.0069	−0.20	−3.75
B, stationary accumulator	3.42	0.0052	−0.15	−2.85

4.2 Distributions



- **Severity (relative speed at impact).** Condition B tracks the reference's tail well and over-produces only the mildest bin; condition C now over-produces *moderate* severities — its fixed 0.5 s response is fast enough that hard crashes are rare but slow enough that many glance draws end in 3–8 m/s impacts.
- **Urgency (t_nr).** Essentially inherited from the seeds by both models, as before.
- **Braking (follower minimum acceleration).** The structural mismatch of the first analysis is largely resolved for condition B: with the no-brake counterfactual and process glances, crashes in which the follower never brakes (glance or no-response covering the conflict) now dominate, matching the reference's non-braking spike. The crash-weighted median follower in B does not brake at all — as in the reference. Condition C retains its hard-braking spikes at the deceleration-bin values.

4.3 Equivalence statistics and the sensitivity ladder

No configuration reaches practical equivalence (ROPE 0.10 for θ; full tables in `replication/causation/out/summary_nb*.md`), but the distance now has a visible structure. Headline θ for P_inj (= relative impact speed; the two give identical bins), Eq. 10 weights, 95% bootstrap HDIs:

configuration	P_inj θ	t_nr θ	a_f,min θ
B, embedded-evasive counterfactual (superseded)	1.44	0.66	1.00
B, no-brake counterfactual, anchored glances	0.87 [0.82, 0.99]	0.40	1.04
B, + process glances (primary)	0.61 [0.55, 0.76]	0.41	1.18
B, + abnormal acceleration	**0.46 [0.41, 0.60]**	**0.30**	1.37
B, process, stationary accumulator	0.42 [0.29, 0.52]	0.39	1.00
C, no-brake, anchored (CBM-native)	**0.40 [0.36, 0.46]**	0.41	1.81
C, + process glances	0.63 [0.44, 0.74]	0.51	1.61
C, + abnormal acceleration	0.59 [0.42, 0.71]	0.45	1.35

Readings:

- **Each methodological correction moved condition B toward the reference**: removing the embedded evasive action (1.44 → 0.87), placing glances as a process (→ 0.61), adding the abnormal-acceleration component (→ 0.46). The remaining distance is dominated by the over-produced mildest-severity bin.
- **The best active-inference configurations now sit at the CBM control's level** (θ ≈ 0.4–0.5 for both) — the response-model gap visible in the first analysis was largely an artifact of the embedded evasive action and the anchor-locked glances. What still separates the two models is *where* they miss: B misses on mild crashes, C on moderate ones and on the braking distribution (a_f,min θ 1.4–1.8 versus B's 1.0–1.4).
- **The abnormal-acceleration component works as intended**: it restores the reference's mean injury risk exactly for B (0.0063 vs 0.0062) by adding the harder, non-braking crashes the four [B24] mechanisms cannot produce. Its cost is a slight overshoot for C.

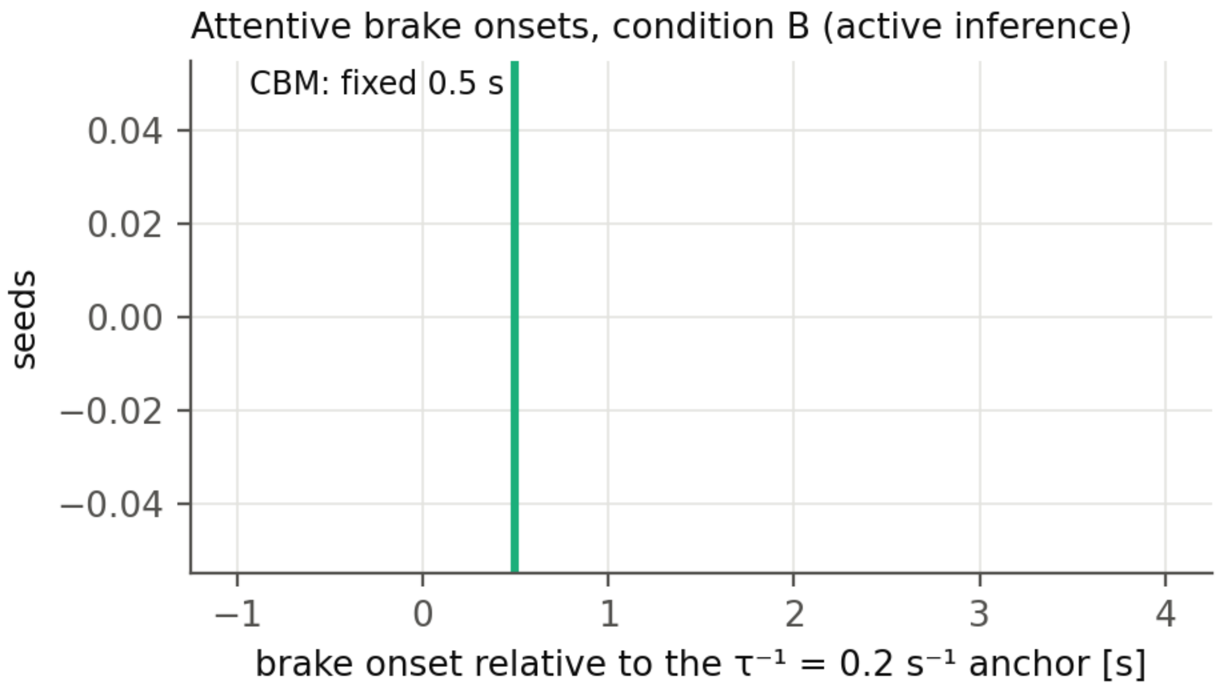
4.4 The counterfactual rule barely matters

Clamping the follower's braking from t = 0 instead of from the lead's onset changes B's P_inj θ from 0.87 to 0.82 and C's from 0.400 to 0.396 (anchored glances) — the choice of clamp rule is not a consequential degree of freedom.

4.5 What the process placement buys, against its cost

For the **active-inference** response, process glances improve severity equivalence substantially (P_inj θ 0.87 → 0.61) besides being theoretically required (the anchored construction assumes the response cannot depend on pre-anchor glance history, which is false for an accumulator). For the **CBM**, the anchored construction — which is exactly Markkula's shortcut for a fixed-delay responder — performs *better* than the process variant (0.40 vs 0.63), partly because the process run's smaller crash-record count (2 076 vs 11 587) widens its uncertainty. Cost: 50 draws per seed versus ~12 bins, minutes either way at tier 1. Conclusion: process placement for active inference, anchored placement for the CBM, each on its own merits; the 50-draw budget is marked for a later increase, which mainly tightens the process runs' HDIs.

5 Response timing



Attentive onsets: active inference median 1.25 s after the anchor (IQR 0.50–1.75 s), CBM fixed at 0.50 s. Ten of 100 seeds respond *before* the anchor — all within 0.45 s of it, all after the lead's braking onset, concentrated at short initial headways — so the active-inference model treats the anchor as nothing special, responding to sub-threshold looming when the gap is short.

The arbiter verdict (2026-08-25). The response-timing claim was initially bracketed by the accumulator's starting convention: zero start (the paper's) versus a "stationary" half-threshold start representing a driver arriving mid-cycle from long steady following; crash probability 0.023 versus 0.004. The tier-2 arbiter run — the full closed-loop model on 23 seeds spanning 1.3–35.5 m/s, four repeats each, lead profiles replayed (`replication/causation/tier2/arbiter_comparison.csv`) — settles it:

comparison (attentive onsets, n = 24 seeds)	median difference	median \	difference\	closer on
closed loop − tier-1 zero start	+0.42 s	0.55 s	**20/24 seeds**	
closed loop − tier-1 stationary start	+1.03 s	1.08 s	4/24 seeds	

The closed loop behaves like the zero start — if anything it responds slightly *later* than the zero-start surrogate, so the stationary correction points the wrong way. The scope of the verdict should be stated precisely: it validates the zero start *for this study's design*, in which simulation windows open a few seconds before the conflict at mostly comfortable headways, so the closed loop's benign drift accumulates little before the event. What a driver carries after minutes of continuous short-gap following is a question neither tier answers, because both open their windows near the conflict; within this design it is a non-issue by construction. Consequently the zero-start results are the study's results, and the conclusion stands: **the active-inference response is slower and more variable than the CBM's fixed rule** — confirmed, not contradicted, by the closed loop.

A deliberate non-action: the +0.30 s median offset between the closed loop and the surrogate could be folded into the surrogate as a calibration correction. We do not do this. Twenty-three seeds are a thin base to fit on, the offset is well inside the response-time dispersion the study is about, and an *untuned* surrogate landing within 0.55 s median absolute difference is a stronger validation statement than a tuned one landing on zero [Opinion].

6 The closed loop is not the CBM with extra steps: the glance-gate finding

The tier-2 adapter (plan §11.4) runs the authors' full closed-loop model on QUADRIS seeds with the lead replaced by a replay of the seed's speed profile, and can force off-road glances through the code's own observation gate (`I_factor`, the mechanism of the Engström et al. 2024 visual time-sharing model).

Forcing a 1.0–3.0 s off-road glance during the conflict did not delay the response. On the test seed the driver braked at 2.4–2.6 s — inside the glance — both under the shipped observation-noise factor (3) and under an effectively total gate (factor 1000). The reason is architectural: the lead's braking had been registered before the glance began, and during the glance the belief cloud coasts forward on its own norm-shaped prediction, so the evidence accumulator keeps filling from *remembered and extrapolated* evidence. Looking away blocks new observations; it does not stop inference.

The CBM assumes the opposite: no information accumulates while the eyes are away, and no response can begin until 0.5 s after they return [B24]. The two architectures therefore make **divergent, testable predictions** that depend on glance timing:

- glance covering the conflict onset → both models delay the response until (or beyond) eyes-return; they roughly agree;
- glance beginning *after* the onset has been registered → the CBM still waits for eyes-return + 0.5 s, while the active-inference driver responds mid-glance on extrapolated evidence.

Naturalistic glance-conditional response data could separate these — drivers who brake while still looking away (or within less than 0.5 s of looking back) in late-glance conflicts would favor the active-inference account. To our knowledge the existing glance-response literature conditions on eyes-off at conflict onset and does not isolate the late-glance case [Opinion]. The tier-1 pipeline mirrors the distinction explicitly: its evidence gate at weight 0 reproduces the CBM assumption, the Svärd et al. partial looming weight is an intermediate, and the tier-2 precision gate is the full active-inference behavior.

7 The pre-selection (exposure) critique, quantified

Applying causation mechanisms to crash-conditioned seeds answers "what does this driver do in situations that crashed for the generator's driver", not "what crashes does this driver produce in traffic". Three numbers from this study's own runs bound how far the two questions are apart:

- **Support loss**: 13 of 100 seeds never crash under the CBM control at any glance or deceleration draw; they drop out of any exposure reweighting entirely.
- **Effective sample size**: reweighting the remaining 87 seeds to exposure (dividing each weight by its crash probability) leaves an effective sample size of **6.9** — three seeds carry the majority of all exposure weight. Exposure-level claims from crash-only seeds would need orders of magnitude more seeds, concentrated exactly in the benign scenarios the crash filter removed.
- **Directional bias**: the missing scenarios are the long-headway, mild-conflict ones in which crashes occur only through glances or non-response — for a glance-causation study, the worst possible region to lose.

The **B-versus-C contrast is far more robust to this critique than any absolute number**, because both response models face the identical conditioned ensemble. Absolute crash rates and severity distributions inherit the full bias. The proper fix is upstream: the pre-filter scenario set (or the fitted scenario-generation models) of [W25a], requested from its author; the 82 real SHRP2 near-crashes in the QUADRIS release are an intermediate exposure extension available now.

8 Conclusions

1. The pipeline now runs the complete design: five switchable causation components, two response processes, digitized real input distributions, a defensible counterfactual, process-placed glances, and the full [W26] readout with the assumption-free severity metric.

2. After the methodological corrections, **the active-inference response is no further from the QUADRIS reference than the CBM control is** (P_inj θ ≈ 0.4–0.5 for both best configurations), and the two miss differently: active inference over-produces the mildest crashes; the CBM over-produces moderate ones and cannot reproduce the reference's non-braking character, which the active-inference conditions now match.
3. The active-inference response-timing claim is **settled by the tier-2 arbiter**: the closed loop matches the zero-start surrogate (median difference +0.42 s, 20/24 seeds closer) and rejects the stationary start, so the active-inference response is slower and more variable than the CBM's rule, with the tier-1 surrogate validated untuned to a median absolute onset difference of 0.55 s across 1.3–35.5 m/s.
4. The architectural finding stands: evidence gating versus inference gating during off-road glances is a real, behaviorally testable difference between the CBM and active inference (section 6).
5. Equivalence in the strict ROPE sense is not reached by any configuration — as it was not by the SCM in [W26] — and section 7's numbers show that exposure-level claims are outside what crash-conditioned seeds can support at any n.

9 Limitations and next steps

- Glance and deceleration distributions are digitized from published figures; the actual SHRP2 bins would remove the residual digitization error and the truncated glance tail. The abnormal-acceleration onset-time distribution of [W25a] is unpublished; the component applies its acceleration from the lead's braking onset instead.
- 50 process draws per seed (marked for increase); the QUADRIS reference is itself model-generated; n = 100 seeds, extendable to the full 5 000 at tier-1 cost.
- The accumulator-initialization question is closed for this design (section 5); what a driver carries into a conflict after minutes of continuous short-gap following remains outside both tiers' windows, and would matter for study designs with long run-ins.
- The stopped/creeping-follower seeds (20 of the 100-seed sample) have now run in tier 2 under the desired-speed convention of 2026-08-25 (the speed the original follower later reached), and the result is a finding rather than a fix: these are **both-stationary queue scenarios** — the lead also stands still — whose original crashes came from the generator's abnormal-acceleration mode driving into the queue (56% of both-stationary cases in [W25a]'s own statistics). The closed-loop driver, correctly, stays put: no attentive response process can produce these crashes, because there is no conflict to respond to. They are reachable only through the abnormal-acceleration and no-response components, which is how the tier-1 pipeline treats them; in the arbiter comparison they contribute no attentive onsets and are excluded automatically.

Regeneration: the tagged commands are in `replication/causation/out/log_nb*.txt`; figures by `replication/causation/make_results_figures.py --tag nbp`; this document by `docs/build_pdf.py docs/crash_causation_results.md`.